

A substance-free autosegmental account of the coronal mutation in English

Abstract:

This paper puts forward a restrictive approach to subsegmental architecture, which assumes that sub-skeletal representations are composed of obligatorily binary branching nodes and substance-free emergent features which serve as pointers to phonetic interpretation. We put the theory to a test by applying its principles to the set of alternations of coronal consonants attested in English before affixes *-ion*, *-ian* and *-ious* in words such as *infectious*, *suggestion*, *revision*, *expansion* etc. It is demonstrated that the requirement for the structure to be binary branching as well as the requirement for English consonants to contain at most three features derive the correct outputs of the coronal mutation. It is also argued that the postulated approach accounts for the behaviour of the items which resist the mutation such as *rebellion*, *Arthurian*, *oblivious* through the last resort operation of node epenthesis.

Key words: substance-free phonology, autosegmental phonology, consonant mutation, features, phonology-phonetics interface

1. Introduction

Similarly to light, which shows the behaviour characteristic of both particles and waves at the same time, phonological representations are both hierarchical and autosegmental in nature. This work appeals to this dual nature of phonological representations and proposes a framework in which phonological structures are composed of strictly hierarchical binary branching nodes, which at the same time may be associated with other nodes in a manner known from the classic works in autosegmental phonology.

In addition, the approach employed below assumes a strict separation of phonology and phonetics. The approaches which postulate that phonology is a formal system independent of phonetic substance (see Hale and Reiss 2008, Odden 2022, Scheer 2022 among others) are obliged to put forward a coherent approach to externalization or phonology-phonetics interface. A natural way of conceiving of such an interface is that it appeals to the operation of transduction in which phonological structure is lexically associated with phonetic information and is re-written as such at the interface (see e.g. Scheer 2022).

This paper postulates that the hierarchical nature of the autosegmental phonological representations plays a crucial role for the transduction. To be precise, we postulate that the phonetic interpretation of the phonological structure proceeds in the bottom-up fashion, commencing at the bottom-most branching node.

The aim of the paper is to apply the general principles of such a framework to the data from English. In particular, we put forward a substance-free autosegmental analysis of the consonant mutation that affects coronal obstruents in the environment of affixes *-ion*, *-ian* and *-ious*. It is assumed that those affixes are lexically equipped with a floating node, which is expected to attach to the structure of the stem-final consonants. The observed mutation is the consequence of the structural modifications that the affected consonants undergo when the floating node is integrated into their representation.

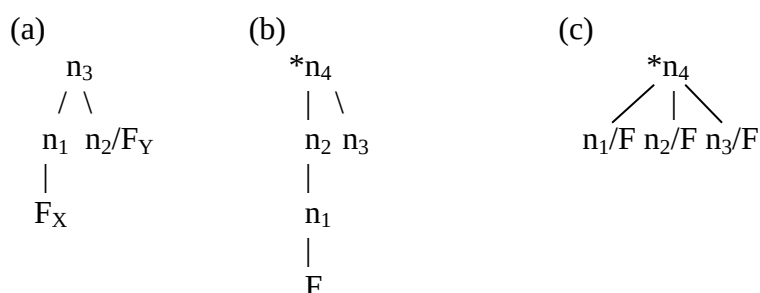
Section 2. introduces the nodes and features as the building blocks of the sub-skeletal phonological structure and discusses the basic well-formedness conditions that a grammatical representation must fulfil. Section 2.1. discusses the workings of phonetic interpretation, while section 2.2. briefly addresses the proposed sources of cross-linguistic differences between sound systems. Section 3. begins with the discussion of the data and introduces the sub-classes of stems affected by coronal mutation. Sections 3.1-3.4. present the analyses of

the stems which undergo the mutation. Section 3.5. provides the analysis of the consonants which systematically resist the mutation. Section 4 concludes the article.

2. A substance-free approach to sub-skeletal architecture: The framework

The framework employed in this article rests on the assumption that the sub-skeletal portion of phonological segments is composed of nodes (n) and features (F). Following Nasukawa (2017, 2018), we assume that the lexical phonological representations are built through the operation Merge, known from the syntactic literature (Chomsky 1995, 2000). As a consequence of Merge, nodes form a binary branching spine. All nodes must be binary branching save the terminal nodes. Features may be attached to terminal or non-terminal nodes. In the latter case they constitute separate branches on their own. The legitimate structure constructed in this fashion is illustrated in (1a). Structures in (1b,c) are impossible.

(1)



As illustrated in (1a), node n_3 may branch into two nodes (n_1 and n_2) or a node (n_1) and a feature (F_Y). Structure (1b) is ungrammatical as node n_2 is a non-branching non-terminal node. Structure (1c) is ungrammatical as it contains a ternary branching node n_4 .

Nodes and features differ in their status in that features, but not the nodes, constitute the instructions for the phonetic realization of the structure (see below for details). While the associations between nodes and features may be removed in the course of the derivation, the associations between nodes are fixed and cannot be removed. At the same time, new associations between nodes may be added to the structure in the course of the derivation.

In the current approach, features are emergent in the sense of Mielke (2008), i.e. there is no universal set of distinctive features inherent in the grammar. Instead, features are postulated at the stage of language acquisition in order to account for the two kinds of phonological behaviour of segments. Firstly, the features account for the existence of lexical contrasts between segments, i.e. segments such as /t/ and /s/ in words such as /kit/ and /kiss/ differ in their featural content as the two words have distinct lexical representations. Secondly, the features are postulated to account for the existence of natural classes of segments which behave in a uniform way with respect to phonological processes or phonotactic generalizations observed in a language. Thus the fact that both /s/ and /t/ give rise to /ʃ/, when subject to the coronal mutation discussed in this paper points to certain common aspects of the featural make-up of /t/ and /s/. At the same time, /s/ and /t/ differ in their featural representation as only the former may be found at the beginning of words before the lateral, cf. *slow* vs. **tlow*.

Features employed in this study are substance-free, unary and privative. The privative nature of phonological features, in which the absence of a feature is opposed to its presence, as well as the assumption that representations are constructed on the basis of their phonological behaviour, results in a radical nonspecification of the phonological representation. The approach postulated here shares this property with approaches such as

Element Theory (Kaye et al. 1985, Harris and Lindsay 1995, Backey 2011 among many others), Government Phonology 2.0 (Pöchtrager 2015), Radical CV Phonology (van der Hulst 2020) or Particle Phonology (Shane 1984). The phonetic detail is not represented phonologically but may be read off phonological representation by the phonetic interpretation procedure, which converts the hierarchical phonological representations into phonetic labels understood by the articulatory-perceptual system.

2.1. Phonetic interpretation of the structure

Apart from the formal requirement that phonological representation be composed of binary branching nodes, a no less important well-formedness condition is that it be phonetically interpretable. Phonetic interpretation is an operation which converts the phonological representation into phonetic labels. Phonetic interpretation (PI) operates in a segment-to-segment fashion, commencing from the right margin of the interpreted linguistic unit (word/morpheme/stem). PI proceeds in a strictly bottom-up order starting with the bottom-most branching node of the representation and supplies phonetic (articulatory and acoustic) labels computable by the phonetic subcomponent of the grammar. The phonetic labels are lexically stored constellations of phonetic information which correspond to phonological nodes and are selected on the basis of the featural content of the relevant nodes. Phonetic interpretation may be sensitive to the presence of a given combinations of features as well as the asymmetrical relation of dominance between the features. This means that representation (2a) is realized as a phonetic label P, while the representation (2b), despite being composed of the same set of features, may be realized as a label P'.



The representations in (2) are supplied with different phonetic content on the basis of two lexically stored interpretational conventions (3a) and (3b).

- (3)
- (a) $\{F_1; F_2\} \leftrightarrow P$
- (b) $\{F_2 > F_1\} \leftrightarrow P'$

The interpretational convention (3a) says that a node under which features F_1 and F_2 co-occur (;), is to be supplied with a phonetic label P. The convention (3b) says that a node under which feature F_2 dominates (>) features F_1 is supplied with the node P'. A feature F_X dominates another feature F_Y if the first node which dominates F_X dominates F_Y . The two conventions are not exactly conflicting as the dominance relation is considered to make (3b) more specific and causes it to be selected over (3a) as a realization of the representation (2b). At the same time, convention (3b) cannot be selected to supply the phonetic label for the representation (2a), which does not contain the dominance relation. Convention (3a), on the

other hand, may realize structure (2b), but only in the absence in the lexicon of the language of a convention (3b).

The higher nodes inherit the phonetic labels supplied for the lower nodes and may contribute additional phonetic information which is combined with the former and send to phonetic execution at the skeletal level represented by nodes C (consonant) or V (vowel).

A phonological structure is phonetically uninterpretable if it does not contain a branching node (4a), if it does not contain an unambiguously lowest branching node (4b) or, if the lexicon of a given language does not contain the interpretational convention that could supply the phonetic label for the relevant node. In the latter case, PI proceeds to the higher node and searches for the available convention to realize this node. If no label is supplied at the level of C/V, the derivation crashes.

(4) Phonetically uninterpretable structures



The structures presented in (4) are universally uninterpretable. The case in which a well-formed structure cannot be interpreted due to the absence of an appropriate interpretational convention is a language-specific situation. Languages may therefore, differ with respect to what structure they are capable of externalizing. The following section discusses further reasons for the observable cross-linguistic variation in phonology.

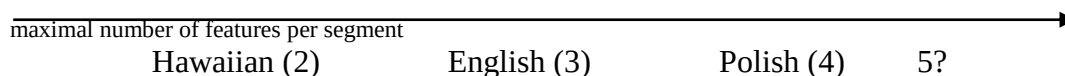
2.2. Why do sound systems differ?

The first answer to the above question has been suggested in the previous section: languages differ because they have different interpretational conventions, which supply different nodes with different phonetic labels. In other words, the first and the major source of cross-linguistic variation in sound systems is that languages have different lexicons. Ideally, the lexicon should be the only source of such a variation. If such a situation obtained, the cross-linguistic variation would boil down to languages differing in their superficial phonetic detail as a result of the existence of the phonology-phonetics interface. The phonological systems of languages would be completely uniform. Unfortunately, such a situation is a naïve idealization. An overwhelming body of cross-linguistic evidence suggests that languages differ in their phonological systems. The aim of phonological theory is to discover and formalize the limits of such a variation in a concise and empirically contentful way.

This paper postulates that much of the cross linguistic variation observed in phonological behaviour of segments boils down to what will be referred to as the Featural Complexity Parameter setting assumed by a given system. Let us assume that languages select a maximal number of features which a segment in a given language may be composed of. The cut-off point at a Featural Complexity hierarchy that a language selects directly affects the possible outputs of phonological processes attested in such a language and, less directly, the number of segments such a language may contain in its inventory. Let us further assume that English allows its consonants to be composed of at most three features. Hawaiian allows its

consonants to contain two features at most, while Polish allows its consonants to contain up to four features. A partial hierarchy of this kind is illustrated in (5).

(5) Featural Complexity Parameters (consonants)



A language which would allow one feature per segment most likely does not exist. Such a language would have trouble forming natural classes of segments. Given the existence of languages which utilize many more consonantal contrasts than Polish, we take it that languages may use more than four features per a segment. The exact empirical consequences of the parametric variation of the postulated form cannot be discussed here. The claim that English does not allow more than three features per one consonant has paramount consequences for the application of coronal mutation in this language.

To conclude the exposition, let us note that the assumptions that constitute the framework presented in this section are not new. The binary branching structures in phonology have been argued for in Pöchtrager (2015), Schwartz (2010), Nasukawa (2017, 2018), van der Hulst (2020) among others. The privative nature of phonological primes have been the cornerstone of many approaches to phonological representations, Element Theory being probably the best-established one. The emergent nature of phonological features has been around at least since Mielke (2008). The claims that phonological representations are free from the phonetic substance has given rise to a growing body of works (see Hale and Riess 2008, Odden 2022). The claim that the interpretational subcomponents of the grammar operate in a bottom-up fashion starting from the lowest branching node is the established view on semantic interpretation (see Heim and Kratzer 1998). In sum, the novelty of the framework presented in this section lies primarily in that it is a new combination of well-known and often well-established claims about how the phonological grammar works. The following sections focuses on how this combination of assumptions may be fruitfully employed to derive a set of relatively complex morphophonological alternations attested in English, i.e the coronal mutation.

3. The analysis of the coronal mutation in English

Although English is much less abundant in morphophonological consonant mutations than, e.g. the system of Polish, some such changes are on the record. The most common and best-analysed mutations attested in English are velar softening (6), t-spirantisation (7), coronal palatalization (8), and coronal mutation (9). They have been analysed in a variety of frameworks in Chomsky and Halle (1968), Rubach (1984), Halle and Mohanan (1985), Kang (2000), Halle (2005), Radomski (2010), among many others.

(6)

- a. *elektri*[k] - *elektri*[s]-ity
- b. *fanati*[k] - *fanati*[s]-ism
- c. *publi*[k] - *publi*[s]-ize
- d. *analo*[g]-ous - *analo*[dʒ]-y
- e. *ri*[g]-our - *ri*[dʒ]-id
- f. *pedago*[g]ue - *pedago*[dʒ]-y

(7)

- a. *pira*/t/e - *pira*/s/-y
- b. *intima*/t/e - *intima*/s/-y
- c. *innocen*/t/ - *innocen*/s/
- d. *pruden*/t/ - *pruden*/s/

(8)

- a. *concep*/t/ - *concep*/tʃ/-ual
- b. *gra*/d/e - *gra*/dʒ/-ual
- c. *u*/z/e - *u*/ʒ/-ual
- d. *sek*/s/ - *sek*/ʃ/-ual

(9)

- a. *elek*/t/ - *elek*/ʃ/-ion
celebra/t/e - *celebra*/ʃ/-ion
infec/t/ - *infec*/ʃ/-ious
educa/t/e - *educa*/ʃ/-ion
- b. *sugges*/t/ - *sugges*/tʃ/-ion
conges/t/ - *conges*/tʃ/-ion
diges/t/ - *diges*/tʃ/-ion
- c. *impre*/s/ - *impre*/ʃ/-ion
aggre/s/-ive - *aggre*/ʃ/-ion
preten/s/e - *preten*/ʃ/-ious
Mar/s/ - *Mar*/ʃ/-ian
- d. *ero*/d/e - *ero*/s/-ive - *ero*/ʒ/-ion
inva/d/e - *inva*/s/-ive - *inva*/ʒ/-ion
conclu/d/e - *conclu*/s/-ive - *conclu*/ʒ/-ion
- d'. *comprehen*/d/ - *comprehen*/s/-ive - *comprehen*/ʃ/-ion
- e. *condescen*/d/ - *condescen*/ʃ/-ion
conten/d/ - *conten*/ʃ/-ious
- f. *rev*/aɪz/e - *rev*/ɪʒ/-ion
dec/aɪd/e - *dec*/ɪʒ/-ion
coll/aɪd/e - *coll*/ɪʒ/-ion
- g. *rebe*/l/ - *rebe*/l(i)j/-on - *rebe*/l(i)j/-ous
victor/r/-y - *victo*/r(i)j/-ous
compa/n/-y - *compa*/nj/-on
en/v/-y - *en*/v(i)j/-ous

The status of velar softening (6) as a phonological change is questionable due to its notorious underapplication (cf. *anar*[k]-ism, *cate*[k]-ize, *monolo*[g]-uist) and possible sensitivity to spelling (only the /k/ spelled as <c> is affected) and will not be analysed here.

While t-spirantization (7) is more regular in that it takes place in numerous abstract nouns, especially those ending in in -y /i/, it admits a large number of exceptions (cf. *calami*/t/-ous - *calami*/t/-y, *current* - *curren*/s/-y but *warran*/t/ - *warran*/t/-y). Due to the existence of those exceptional cases we will leave the status of t-spirantization for further research.

Coronal palatalization applies regularly before the affix -ual /*(ju)əl*/. In the case of some items it is optional and register-dependent (see *seks*/juəl/ vs. *sek*/ʃuəl/) in others, e.g. *usual*, it is obligatory.

Given the space limitations, this paper focuses only on the descendants of Early Modern English palatalization illustrated in (9). These changes are jointly referred to as the *coronal*

mutation. The relevant changes affect alveolar obstruents obligatorily before the affixes *-ion*, /*(ij)ən*/ and *-ious* /*(ij)əs*/ and often variably before affix *-ian* /*(ij)ən*/. They cause /t/ to surface as post-alveolar /ʃ/ (9a) unless it is preceded by /s/, in which case the post-alveolar affricate /tʃ/ surfaces (9b). The items in (9c) illustrate the cases in which the underlying /s/ surfaces as /ʃ/. The stems in (9d) and (9e) surface with the lenis post-alveolar spirant /ʒ/ or the fortis spirant /ʃ/, if preceded by the nasal. The items in (9d), surface with stem-final /d/ if they are verbs, and with /s/ in front of the affix *-ive* /*iv*/.

In the items such as (9f), the consonant mutation is accompanied by a vowel mutation. Since a reasonably constrained model of phonology cannot derive a pattern in which the diphthong /aɪ/ is shortened to /ɪ/ before *alternating* /z/, we will disregard vowel alternations and focus only of the consonantal portion of the alternation stems, i.e. the derivation of /ʒ/ from /z/.

Segments other than coronal obstruents resist the mutation (9g) and the stem-final segments in relevant words are followed by a vowel /i/ or a semi-vowel /j/.

To begin with, let us present preliminary lexical representations of the segments which constitute the input and the output of the mutation, i.e. the coronal obstruents found in English (see 10).

(10) Coronal obstruents in English

a)	b)	c)	d)
/t/	/d/	/tʃ/	/dʒ/
/s/	/z/	/ʃ/	/ʒ/
C	C	C	C
n	n	n	n
\	\	\	\
n F ₅	n F ₄	n F ₅	n F ₄
\		\	
n F ₄	F ₂	n F ₄	F ₃
F ₂			

The representations found in (10) are to be treated as the default configurations which render the relevant phonetic objects. The discussion in the sections to follow will reveal that certain representations may be richer, i.e. contain more nodes and features, than what (10) illustrates and that certain nodes may be rearranged with respect to other nodes. All such manipulations are legitimate so long as the representations conform to the general principles discussed in section 2: the binary-branching requirement, Featural Complexity Constraints and the need for the representations to be phonetically interpretable.

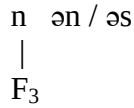
The interpretational conventions, which convert the representations of English coronal obstruents into their phonetic realizations are presented in (11).

(11)

- (a) {F₅} ↔ ‘fortes obstruent’
- (b) {F₄} ↔ ‘occlusion’
- (c) {F₄;F₃} ↔ ‘post-alveolar non-continuant’
- (d) {F₂;F₃} ↔ ‘post-alveolar’
- (e) {F₂;F₄} ↔ ‘alveolar non-continuant’
- (f) {F₂;F₅} ↔ ‘fortes alveolar obstruent’

Let us further assume, that the affixes before which coronal mutation takes place are lexically represented with a floating node hosting feature F_3 , which anchors onto the preceding linearized material triggering the relevant mutations (see 12).

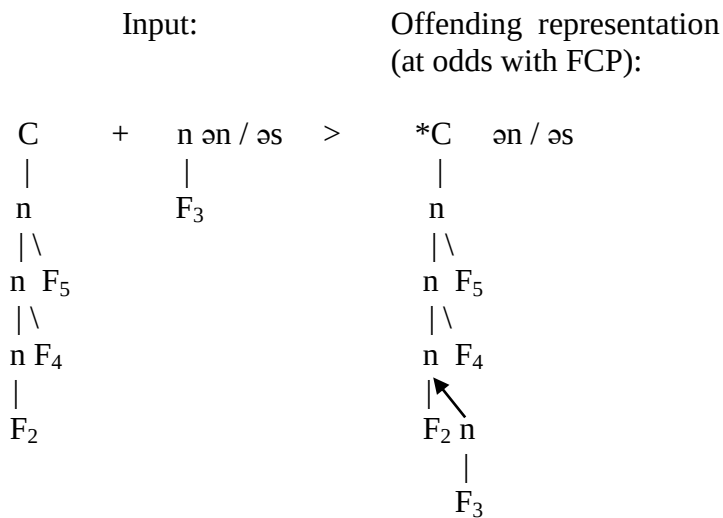
(12) The representation of affixes *-ion -ian* /ən/, *-ious* /əs/



3.1. The derivation of stems in /t/ and /s/ (*obligation, suggestion, impression*)

Let us propose that the floating node anchors onto the stem-final consonant¹ as low as possible without violating the universal and language-specific constraints discussed in section 2. When one of the affixes found in (12) is concatenated with stems which lexically terminate in the fortis alveolar plosive /t/ (10a) the floating node attempts to anchor as the sister to feature F_2 . The resulting structure, however, is at odds with the Featural Complexity Parameter setting for English, according to which English does not tolerate consonants specified for more than three features (see 13).

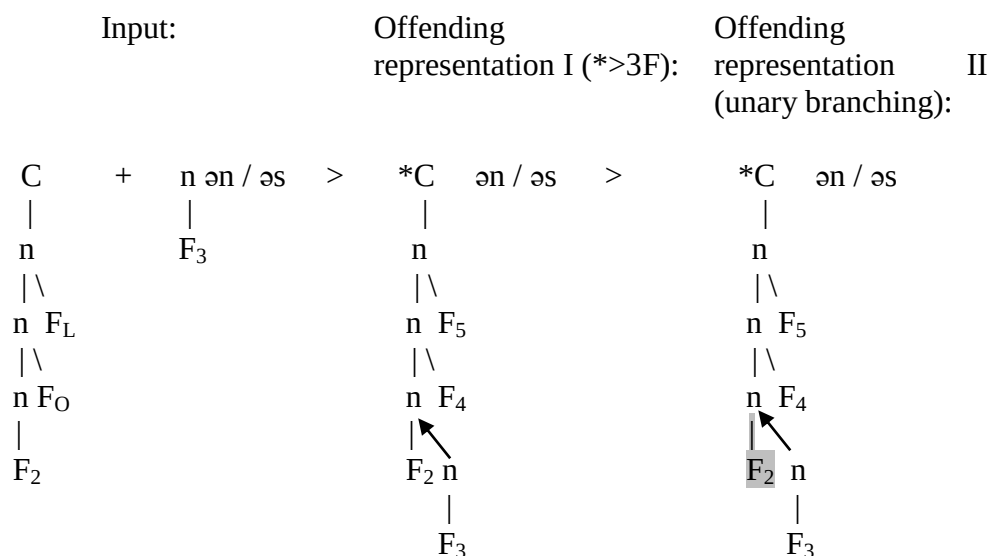
(13) Offending structure I



Note that the repair operation, which results in the delinking and deletion of feature F_2 is not an option as it results in the derivation of the unary branching structure (14) in which the sister to feature F_4 dominates only one node.

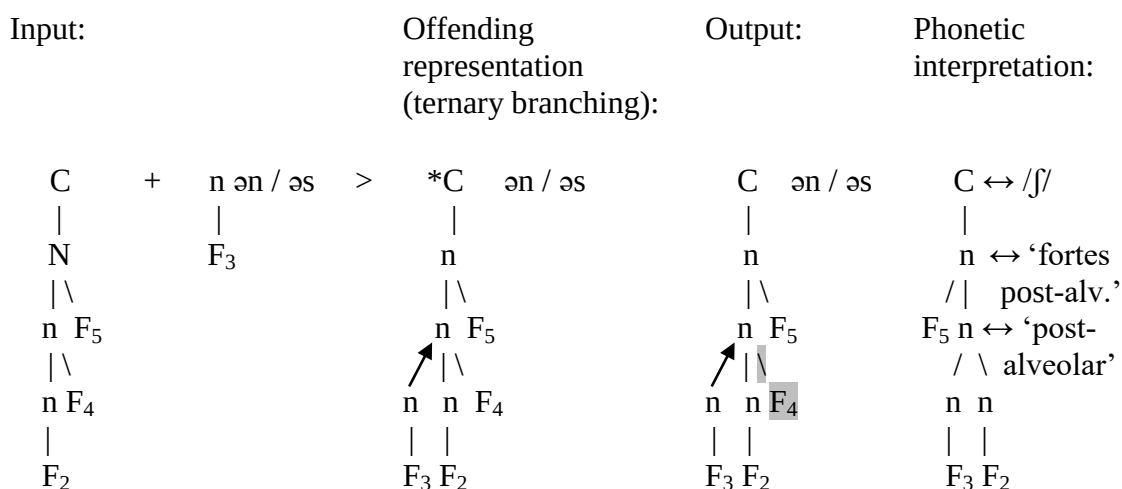
¹ See van Oostendorp (2007) and Trommer (2011) for proposals as to why floating autosegments favour heteromorphemic over tautomorphemic anchoring.

(14) Offending structure II (derivation of a unary branching representation)



Since it is not possible for the floating node to anchor as the sister of feature F_2 , it attempts to anchor as the sister of feature F_4 . This results in the illegal ternary branching structure, which may, however be repaired by the deletion of feature F_4 . This is illustrated in (15).

(15) Legitimate derivation /t/ > /ʃ/

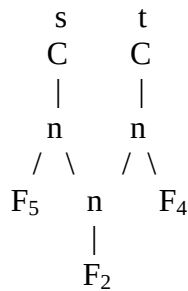


The phonetic interpretation of the output structure in (15) commences by establishing the exponent of the bottom-most branching node. The convention which realizes the features dominated by this node are is (11d, $\{\text{F}_2; \text{F}_3\} \leftrightarrow \text{'post-alveolar'}$). The relevant node is interpreted as 'post-alveolar'. The next branching node in the structure inherits the label 'post-alveolar' and contributes the label 'fortes obstruent', by convention (11a). The representation is realized as the fortes post-alveolar fricative /ʃ/.

In sum, the derivation described above causes the stems which lexically terminate in the alveolar plosive /t/ (*celebra*/t/, *infec*/t/ etc.) to surface with the post-alveolar spirants in words such as *celebta*/ʃən/ and *infec*/ʃəs/, i.e. when concatenated with affixes *-ion* and *-ious*.

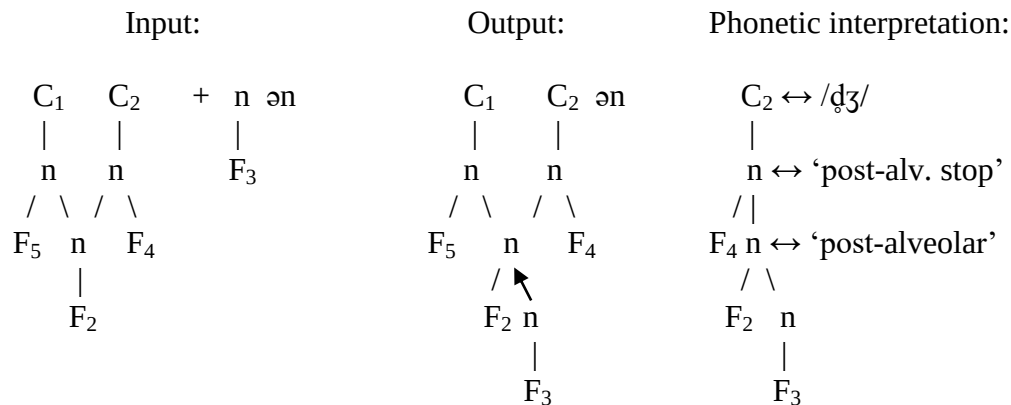
The derivation proceeds in a minimally different way if the alveolar plosive is preceded by the fricative /s/ (10c). It is a well-established fact about English, that the contrast between the lenes and fortes plosives is neutralized after the consonant /s/. In the current framework, this means that the plosives which follow /s/ are represented without the structural layer containing feature F₅. The /st/ sequence found in stems such as *sugge/st/* is represented as in (16), i.e. as a partial geminate in which the fricative and the stop share node hosting feature F₂.

(16) The representation of the sequence /st/



The concatenation of the relevant stems with the affixes equipped with the floating node is illustrated in (17).

(17) The concatenation of stems in /st/ + *ion* (e.g. *suggest-ion*)

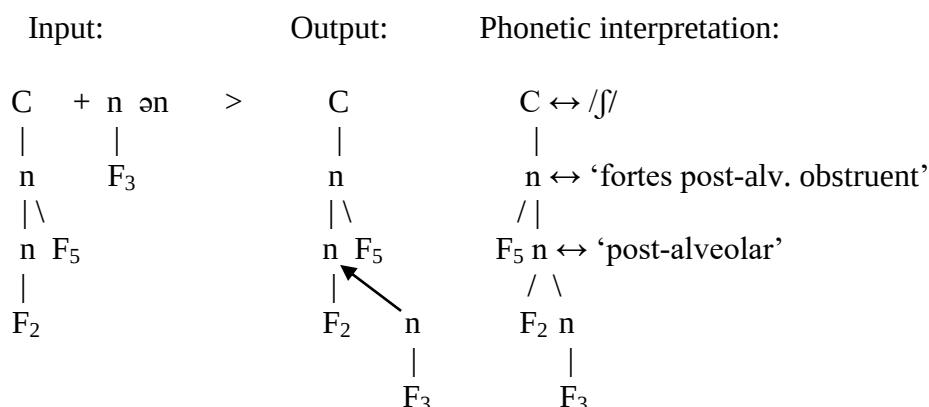


In this case, the bottom-most branching node is again interpreted by convention (11d, {F₂;F₃} ↔ 'post-alveolar'). The feature attached to the higher branching node further contributes the label 'occlusion' and the output is realized as the post-alveolar non-continuant, i.e. a non-fortes /tʃ/ or a voiceless version of the /dʒ/.²

The concatenation of the stems which terminate in the spirant /s/ (10c) in the lexical representation with the relevant affixes is illustrated in (18).

² In careful speech, the initial sound of the cluster, i.e. /s/, may resist assimilation and be pronounced with the alveolar place of articulation. In such cases, an alternative representation is proposed in which /s/ and /t/ do not share the node hosting the feature F₂.

(18) The concatenation of stems in /s/ + *ion* (e.g. *agress-ion*)



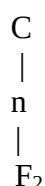
The output is realized as /f/ by virtue of conventions (11d, {F₂;F₃} ↔ ‘post-alveolar’) and (11a, {F₅} ↔ ‘fortes obstruent’).³

3.2. The triplets in /d/, /s/ and /ʒ/ (*conclude* - *conclusive* - *conclusion*)

As noted above, the stems found in (9d) give rise to triplets terminating in /d/, if they are verbs, /s/ if they are followed by the affix *-ive* and the post-alveolar spirant in the presence of the affix *-ion*. The *comprehend/expand* subclass (Xd') constitutes a somewhat different set of cases, as they surface with /ʃ/ before the affix *-ion*. This subclass will be analysed in the following subsection.

Let us postulate that stems such as *conclude* terminate in an abstract, i.e. phonetically uninterpretable, ‘coronal’ consonant specified only for feature F₂ (see 19).

(19) The final segment in stems such as *conclude* - *conclusive* - *conclusion*

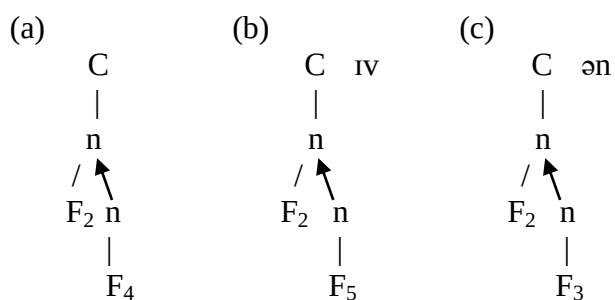


Let us further assume that morphosyntactic head *v*, which categorizes the class of stems including *conclude*, *comprehend*, *expand* etc. and contributes to their interpretation as verbs has the floating node hosting feature F₄ as its sole phonological exponent. At the same time, affix *-ive*, categorizing the relevant stems as adjectives, is lexically represented with a floating node and feature F₅.⁴ When concatenated with the stems which terminate in the consonant represented as (19), the floating nodes anchor as sisters to feature F₂ (see 20).

³ The derivation of the pair *confu/z/e* - *confu/ʒ/-ion* proceeds in an analogous fashion with the output interpreted as lenes due to the absence in the structure of the feature {F₅}.

⁴ In fact, we have found only one stem which features a lenes obstruent and affix *-ive*. This is the stem *gerund*, which gives rise to the word *gerundive*. This word clearly differs semantically from most other words featuring the affix *-ive* and its status as an adjective is questionable. Given these facts, one might be inclined to assume that the affix *-ive* attested in *gerundive* is a different affix that the one attested in items such as *conslusive*, *extensive*, *investigative* etc.

(20) The derivation of items such as *clonclu*/d/e, *conclui*/s/-ive and *conclu*/ʒ/-ion



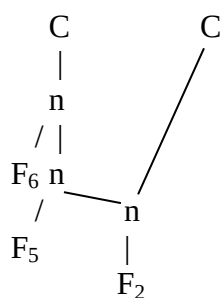
The branching nodes in (20a) and (20b) are interpreted by virtue of specialized conventions (11e, $\{F_2; F_4\} \leftrightarrow$ ‘alveolar non-continuant’) and (11f, $\{F_2; F_4\} \leftrightarrow$ ‘fortes alveolar obstruent’). Ultimately, they are realized as /d/ and /s/, respectively. The representation in (20c) is supplied with phonetic substance by convention (11d, $\{F_2; F_3\} \leftrightarrow$ ‘post-alveolar’) and pronounced as /ʒ/.

3.3. The stems in -nC (*comprehend/expand*, *condescend*)

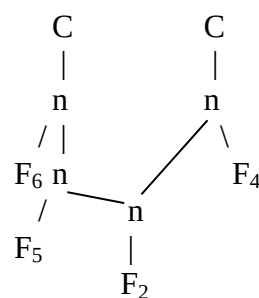
This section discusses the derivation of the stems which terminate in a cluster of a nasal consonant and the lenis alveolar obstruent, i.e. classes (9d') and (9e). In both those classes the cluster is obligatorily homorganic. This fact is represented structurally as the nasal consonant and the following obstruent sharing the node which hosts feature F_2 (see 21).

(21)

(a) *comprehend/expand*-class



(b) *condescend*-class

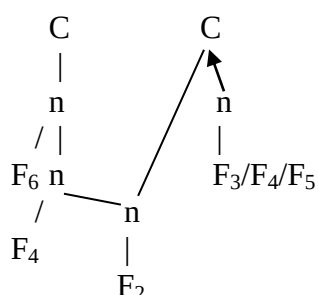


Feature F_6 contributes the property of nasality by virtue of the interpretational convention $\{F_6\} \leftrightarrow$ ‘nasal’.⁵ Importantly, the Featural Complexity Parameter setting for English prevent floating nodes from docking as the sister to feature F_2 , as that would result in the nasals hosting more than 3 features. Thus, the floating nodes are forced to anchor onto the higher nodes. In the *comprehend/expand* class (21a), this is node C. The anchoring results in the representation (22a) and the phonetic interpretation takes place at node C. In the *condescend* class, the floating node docks onto the node hosting feature F_4 . The phonological grammar avoids the derivation of the ternary branching structure by removing feature F_4 (see 22b).

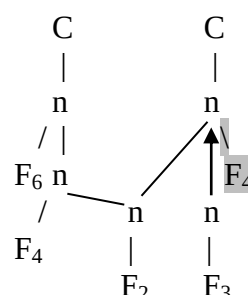
⁵ In (21), the representation of nasals are structurally simplified for clarity of exposition. We present the complete structure of nasals in (27) in section 3.5.

(22)

a) *comprehend/expand-class*



b) *condescend-class*

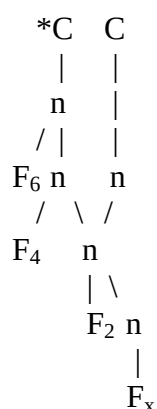


Given the interpretational conventions we have been working with so far, the structures featuring F_3 in (22) should be realized as / ʒ /. This is the case, as the only way for a structure to be interpreted as a ‘fortes obstruent’ is for it to contain feature F_5 . However, such a result is incorrect. As a matter of fact, the contrast between the lenes / ʒ / and fortes / ʃ / is neutralized in favour of the latter after / n / in English. English does not feature clusters such as / nʒ /.

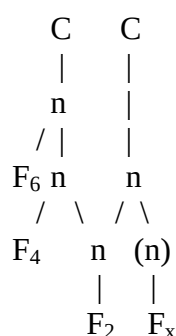
Let us propose that the absence of such partially-homorganic nasal-lenens fricative clusters is a consequence of two factors. The first of them is the Featural Complexity Parameter setting, which says that English sets a limit to the number of features in consonants: a consonant cannot contain more than three features. As a consequence, if a following consonant has a complex featural specification, i.e. if there is more than one feature which occupy the bottom-most branching node, the features which define the nasal cannot dominate two such features. There are two series of affricates in English which have complex place specification. These are dental affricates / tθ / and / dð / and the post-alveolars. Thus a cluster composed of a nasal and post-alveolar/dental fricative is not represented as in (23a). Instead, it must be represented as in (23b).

(23)

(a) illegal n+C cluster



(b) legitimate n+C cluster



In sum, in the case of complex specification, a nasal and a following segment may only form a partial geminate, never a full geminate.

Let us further propose that the presence of the fortes realization of the relevant fricatives after nasals is a consequence of the availability of two interpretational conventions illustrated in (24).

(24)

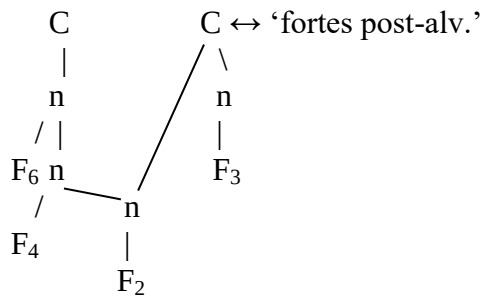
(a) $\{F_2 > F_3\} \leftrightarrow$ ‘post-alveolar’

(b) $\{F_2; F_3\} \leftrightarrow$ ‘fortes post-alveolar’

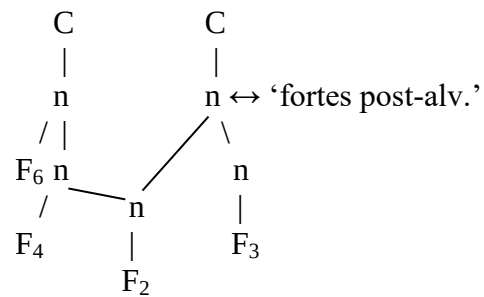
Convention (24a) says that a node which hosts a dominance relation between features F_2 and F_3 contributes the label ‘post-alveolar’ to the phonetic interpretation of the structure. According to (24b), the node under which features F_2 and F_3 simply co-occur, contributes the label ‘fortes post-alveolar’.⁶ Since the structures in (22), do not contain the asymmetrical relation of dominance between features F_2 and F_3 , it is the convention (24b) which supplies the phonetic labels for the relevant nodes. This is illustrated for the two sub-classes in (25).

(25)

(a) *comprehens-ion/expans-ion-class*



(b) *condescens-ion-class*



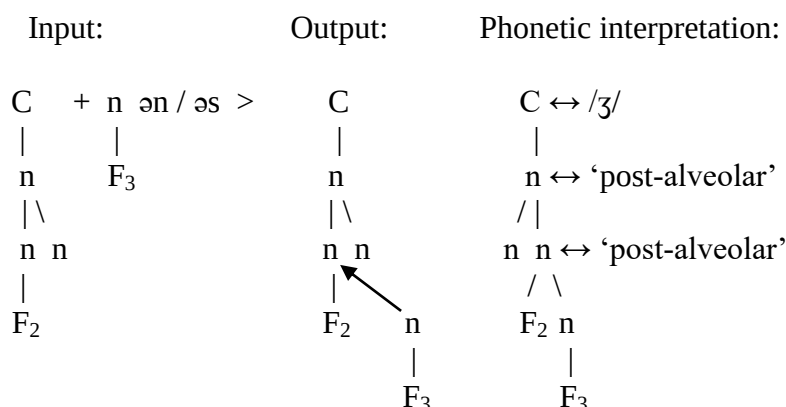
In sum, in items such as *comprehend/expand* and *condescend* the final consonant mutates into the fortes fricative because the floating node cannot anchor in a manner that would contribute an additional feature to the structure of the nasal as such a structure would contain four features. Due to the availability of conventions (24), the resulting representation are interpreted as fortes.

3.4. The *revise* - *revision* class

On the assumption that the stems from the *revision*-class terminate in an underlying /z/, the integration of the floating node proceeds as presented in (26)

⁶ Since dental fricatives do not participate in the relevant mutations, we postulate a convention $\{F_1 > F_2\}$ to contribute the label ‘dental’ in both lenes /ð/ and fortes /θ/. The latter additionally contains feature F_5 , interpreted as ‘fortes obstruent’.

(26)

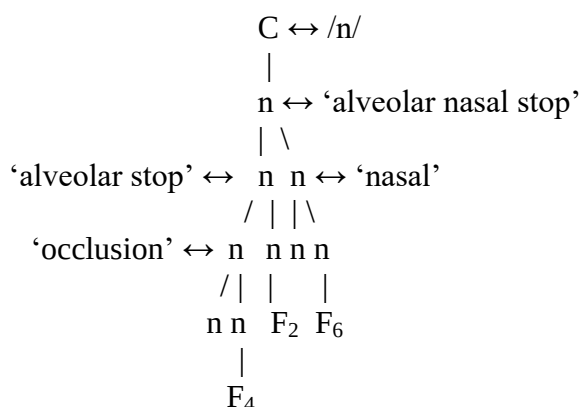


The floating node anchors as the sister to feature F_2 . The resulting branching node is interpreted as post-alveolar by convention (24a). The relevant consonants are realized as the spirant /3/.

3.5. Segments which resist the mutation

The stems which resist mutation in the context of affixes *-ion*, *-ian* and *-ious* do if for several reasons. The nasal /n/ attested in the item *compan-ion* is structurally too large to be enriched with another feature. Let us postulate that the structure of the nasal is as presented in (27).

(27) The representation of /n/ in English



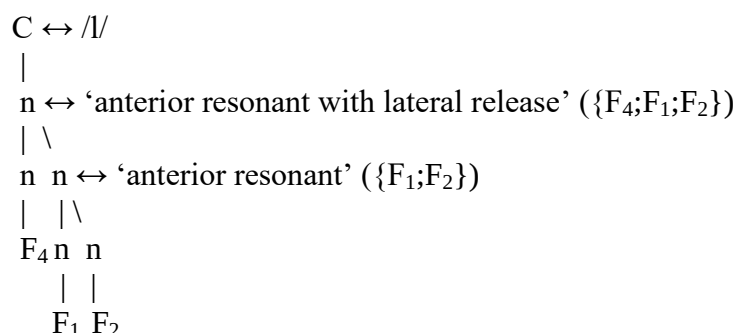
The structure presented in (27) is a universal structure for an alternation-resistant segment. The augmentation of any of the layers of the structure in (27) with an additional node would result in the violation of the core principles of the phonological grammar or is otherwise blocked by the Featural Complexity Parameter setting of English.

The lateral (28) is also structurally complex and resists enrichment with an autosegmental node.⁷ The representation of the lateral is designed to account for the well-attested stages of the weakening of /l/ in English: darkening to /ɫ/ and vocalization to /w/. The former is analysed as the detachment of feature F_2 , while the latter is expressed as a further removal of feature F_4 .

⁷ Interestingly, like nasals, the lateral may be followed only by fortes post-alveolar (and dental) fricatives. This lends credibility to the analysis of laterals as structurally rich.

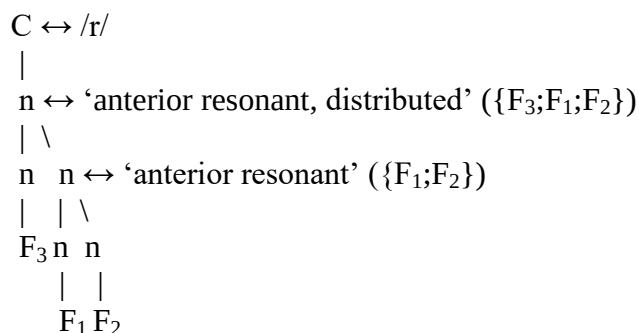
As in the case of the nasals, a floating node cannot be integrated with the structure (28) without the violation of the general well-formedness conditions.

(28)



The rhotic, presented in (29), resist the mutation due to its structural complexity and the fact that it already contain feature F_3 in its subsegmental representation.

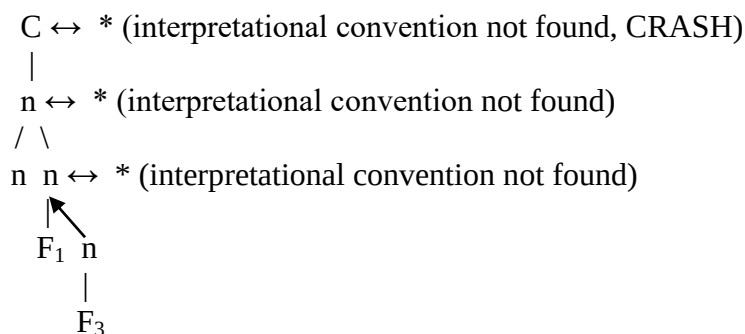
(29)



The presence of F_3 in the rhotic is justified given that, in many varieties of English, coronal obstruents tend to be pronounced with an alveo-palatal place, when followed by /r/, cf. *misery* rendered as *mi/ʒr/y*, *grocery* as *gro/ʃr/y* (see Wells 2008). The presence of feature F_1 , which contributes the 'rounded/labial' quality in the phonetic interpretation of the English rhotic is also manifested in the varieties of South-East London English, which realize the rhotic as the labio-dental approximant /v/ (see Tollfree 1999: 174).

The stems such as *envious* /env(i)jəs/ and *oblivion* /ɒblɪv(i)jən/ resist mutation for a different reason. In those items, the anchoring of the floating node hosting feature F_3 gives rise to a structure which cannot be interpreted phonetically due to the fact that English does not have an interpretational convention that would realize the output.

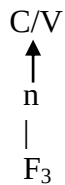
(30) The anchoring of the floating node onto /v/



Since a structure in which feature F_1 dominates or co-occurs with feature F_3 is not found in the English lexicon, the resulting structure cannot be interpreted at the phonology-phonetics interface, and the derivation crashes. The possibility of the derivation crashing due to the unavailability of the interpretational convention is a consequence of the minimalist assumption that the (phonological) grammar exists as an optimal means of meeting the interface condition of interpretability.

In the floating node cannot anchor onto the preceding stem-final obstruent, it is realized as a vowel /i/ or an approximant /j/, giving rise to forms such as *rebe/liən/~/ljən/*, *Arthu/riən/~/rjən/*, *ceremo/niəs/~/njəs/* etc. We propose to account for these realization as instantiating the last resort strategy, which involves an epenthesis of a node. Node epenthesis allows the floating structure to form a branching node, which may be subject to phonetic interpretation. Note that if a floating node carrying a feature simply anchors onto a skeletal position (31), the resulting structure is not phonetically interpretable, due to an unavailability of a branching node.

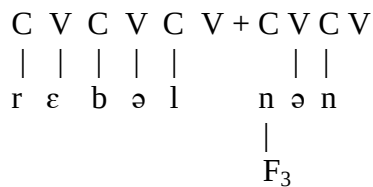
(31) A phonetically uninterpretable structure



The current analysis follows the core assumptions of the approach to syllabic structure formulated in Lowenstamm (1996), Scheer (2004) and Cyran (2010) among others. The framework known as Strict CV Phonology, assumes that CV is the only syllable type and all languages represent words using the sequences of C positions followed by V positions, which give rise to consonants and vowels, respectively. As a consequence, words which terminate in consonants do so only superficially, and at the syllabic or skeletal level they terminate in empty V positions, i.e. V positions devoid of subsegmental structure. Similarly, vowel-initial words in fact begin in empty C positions.

Given these general assumptions the stem *rebel* terminates in an empty V and the affix *-ion*, begins with an empty C. The concatenation (+) of these two morphemes is represented as in (32).

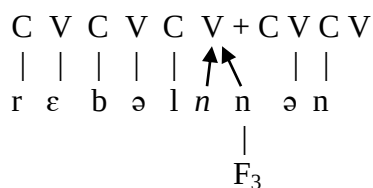
(32)



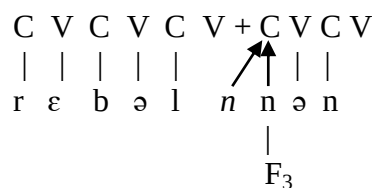
Since the floating node cannot anchor onto the /l/, as the latter is structurally too complex, the grammar invokes a last resort operation of node epenthesis, which allows the floating structure to be realized on the empty V, which is pronounced as /i/. Alternatively, the floating node may anchor onto the affix-initial C position, which, in such a case, is realized as /j/. The two representational options are illustrated in (33). The epenthetic nodes are italicized.

(33)

(a) *rebe/liən/*



(b) *rebe/ljən/*



In sum, the stems which terminate in the nasal, the lateral and the rhotic resist mutation as they are already composed of the maximal number of three elements and their structures cannot be augmented with a floating node without violating a general well-formedness principles. The stems in /v/ may in principle host the floating node. Nevertheless, the absence of the interpretational convention that would supply the phonetic label to the combination of features F_1 and F_3 makes it impossible for the floating node to dock onto the labiovelar fricative.

In those cases, the grammar employs the last resort strategy of node epenthesis, which allows the floating node to dock onto the empty syllabic positions deriving the representations with vowel /i/ or consonant /j/.

4. Conclusion

This paper demonstrated that a substance-free autosegmental framework, which employs privative features can successfully account for the complex and opaque pattern of coronal mutation in English. The attested outputs of the mutation are a function of several factors. The first of them is the size of the floating material that anchors onto the stem-final segments to provoke the mutation. In the case of coronal mutation this is a node and a feature. In the case of the palatalization attested before affix *-ual* (see 8) the floater may well be just the feature F_3 . The second factor is the lexical representation of the stem-final consonants. If the target consonant is structurally too complex the anchoring may require adjustments, as in many cases of the underlying /t/. If the lexical representation makes no available place for the anchoring of a node (see 27, 28), or if they already contain an instance of the anchoring feature, the docking may be blocked. Finally, the docking may be blocked if it gives rise to a phonetically uninterpretable consonant. In English the combination of the features F_1 , which typically contributes ‘labial/rounded/anterior non-coronal/low F_1 and F_2 ’ phonetic label and F_3 (‘post-alveolar/palatal/distributed/high F_2 ’) does not correspond to any interpretational convention. In other languages, however, the combination of similarly interpreted features may be realized as a palato-labial consonant (Polish) or a front-rounded vocoid (French).

According to the approach outlined in this contribution, sound systems differ in the availability of particular interpretational conventions and the settings of the Featural Complexity Parameter. The exact limits that these two factors set on the possible shapes of sound systems seems to be an exciting and promising direction of research.

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